

Complete Variable List for Euroleague Clustering Analysis

All 23 Variables (European Metrics)

This list matches the NBA paper's Table 1, adapted for Euroleague with native European measurements.

1. Physical Attributes

#	Variable Name	Unit	Description	Data Source
1	Height_cm	cm	Player height in centimeters	PlayerStats API

2. Rebounding Metrics

#	Variable Name	Unit	Description	Formula/Source
2	OReb_Rate	%	Offensive rebound rate	OReb / TotalReb
3	DReb_Rate	%	Defensive rebound rate	DReb / TotalReb

3. Playmaking & Defense

#	Variable Name	Unit	Description	Formula/Source
4	Assist_Rate	ratio	Assists per field goal made	AST / FGM
5	Steal_Rate	per game	Steals per game	Direct from API
6	Block_Rate	per game	Blocks per game	Direct from API
7	Turnover_Rate	per 100	Turnovers per 100 possessions	Per100Possessions API

4. Scoring Volume & Efficiency

#	Variable Name	Unit	Description	Data Source
8	Points_Per100	per 100	Points per 100 possessions	Per100Possessions API
9	Usage_Rate	%	Share of team possessions used	$(FGA + 0.44 \times FTA + TO) / GP$
10	Player_Efficiency_Rating	PIR	Euroleague efficiency metric	PIR from API

5. Free Throw Metrics

#	Variable Name	Unit	Description	Formula/Source
11	FTr	ratio	Free throw rate	FTM / FGA
12	FT_Pct	%	Free throw percentage	$FTM / FTA \times 100$

6. Field Goal Volume & Accuracy

#	Variable Name	Unit	Description	Data Source
13	FGA_Per100	per 100	Field goals attempted per 100 poss	Per100Possessions API
14	2FG_Pct	%	Two-point field goal percentage	$2FGM / 2FGA \times 100$
15	3FG_Pct	%	Three-point field goal percentage	$3FGM / 3FGA \times 100$

7. Shot Assist Rates

#	Variable Name	Unit	Description	Data Source
16	2FG_Assist_Rate	%	% of 2PT shots assisted	Shot tracking data
17	3FG_Assist_Rate	%	% of 3PT shots assisted	Shot tracking data

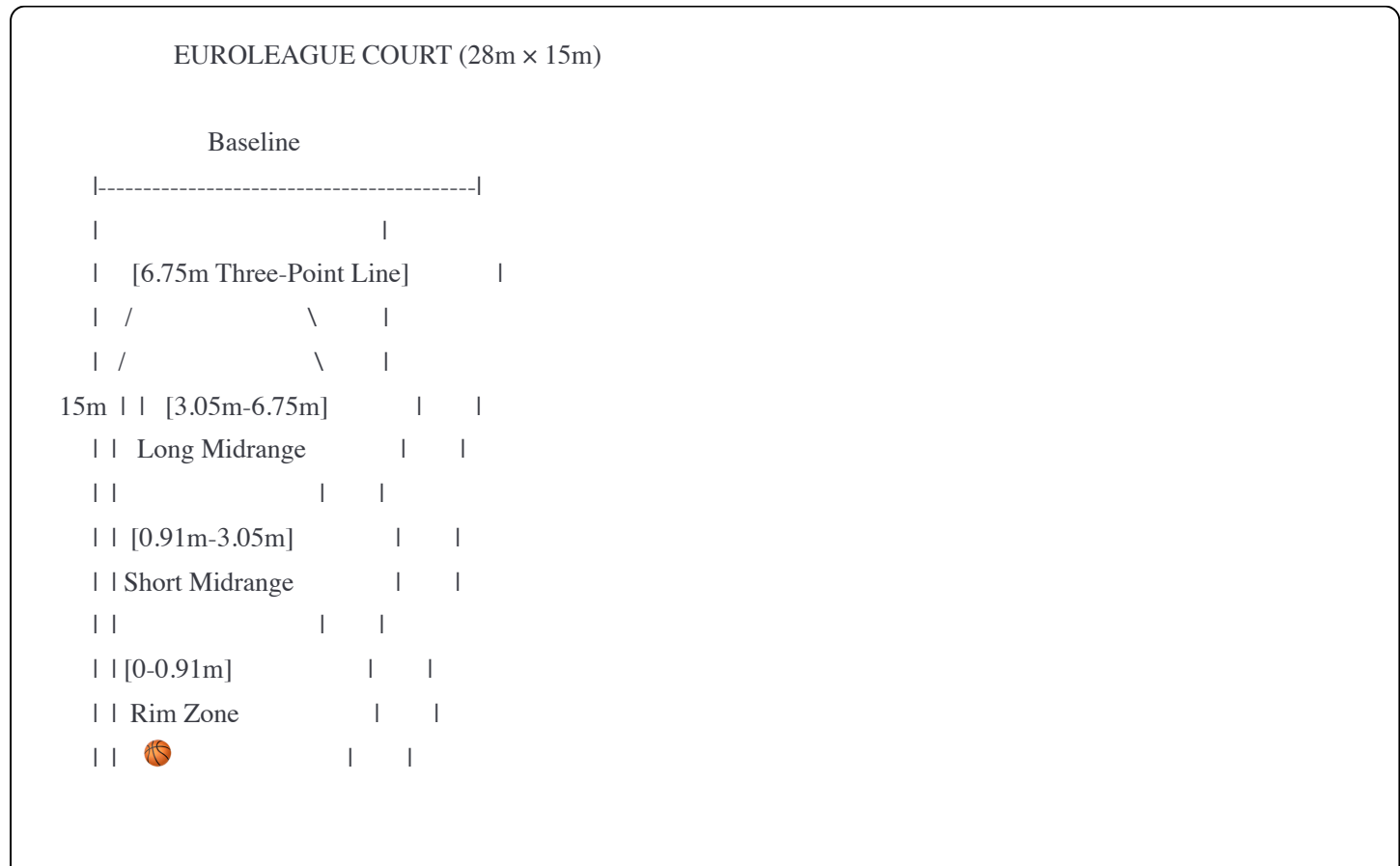
8. Shot Selection

#	Variable Name	Unit	Description	Formula/Source
18	3FGA_Pct	%	% of FGA that are 3-pointers	$3FGA / FGA \times 100$
19	Corner_3FGA_Pct	%	% of FGA from corner 3	Shot location data

9. Shot Location Distribution (European Metrics)

#	Variable Name	Unit	Zone Description	Equivalent NBA Zone
20	Dunk_Attempt_Rate	%	% of FGA that are dunks	Same
21	0-0.91m_FGA_Pct	%	At rim (0-0.91 meters)	0-3 feet
22	0.91-3.05m_FGA_Pct	%	Short midrange (0.91-3.05m)	3-10 feet
23	3.05m-3p_FGA_Pct	%	Long midrange (3.05m-6.75m)	10ft-3point

Quick Visual: Shot Zones on Euroleague Court



28m Court Length

Key Distances:

- **0-0.91m:** Paint/Rim zone (dunks, layups)
- **0.91-3.05m:** Short midrange (floaters, short jumpers)
- **3.05m-6.75m:** Long midrange (elbow to 3-point line)
- **6.75m+:** Three-point territory

Data Collection Checklist

When collecting Euroleague data, ensure you have:

✓ Core Statistics (Available from API)

- ☒ Height in **centimeters**
- ☒ Rebounds (Offensive, Defensive, Total)
- ☒ Assists, Steals, Blocks, Turnovers
- ☒ Points, FGA, FGM (per game and per 100)
- ☒ 2PT and 3PT shooting (attempts, makes, percentages)
- ☒ Free throws (attempts, makes, percentage)
- ☒ PIR (Performance Index Rating)

✓ Calculated Metrics

- ☒ Rebound rates (OREb%, DReb%)
- ☒ Assist rate (AST/FGM)
- ☒ Usage rate approximation
- ☒ Free throw rate (FTM/FGA)
- ☒ 3-point attempt percentage

✓ Shot Location Data (Requires Shot Tracking)

- ☒ Corner 3-point attempts
- ☒ Dunk attempts
- ☒ Shot distances in **meters**
- ☒ Zone classifications (0-0.91m, 0.91-3.05m, 3.05m-6.75m)
- ☒ Assist tracking for 2PT and 3PT

Sample Data Row

Example player: **Jan Vesely (2023-2024 Season)**

```
python

{
  # Physical
  'Height_cm': 213,

  # Rebounding
  'OREb_Rate': 0.18,
  'DReb_Rate': 0.25,

  # Playmaking & Defense
  'Assist_Rate': 0.32,
  'Steal_Rate': 0.8,
  'Block_Rate': 1.2,
  'Turnover_Rate': 1.5,

  # Scoring
  'Points_Per100': 22.5,
  'Usage_Rate': 18.3,
  'Player_Efficiency_Rating': 19.8,

  # Free Throws
  'FTr': 0.45,
  'FT_Pct': 72.3,

  # Field Goals
  'FGA_Per100': 15.2,
  '2FG_Pct': 68.5,
  '3FG_Pct': 0.0,

  # Shot Selection
  '2FG_Assist_Rate': 85.0,
  '3FG_Assist_Rate': 0.0,
  '3FGA_Pct': 0.0,
  'Corner_3FGA_Pct': 0.0,

  # Shot Zones (Metric)
  'Dunk_Attempt_Rate': 35.2,
  '0-0.91m_FGA_Pct': 82.5, # Most shots at rim
  '0.91-3.05m_FGA_Pct': 15.0, # Some short midrange
  '3.05m-3p_FGA_Pct': 2.5 # Rarely shoots long
}
```

Profile: Traditional center who operates exclusively in the paint, doesn't shoot 3s, high assist rate on 2PT shots (alley-oops), elite finisher at rim.

Units Summary Table

Variable Type	Unit	Example Value	Notes
Height	cm	203	No conversion needed
Rates	ratio	0.15	Decimal (15%)
Percentages	%	38.5	Already multiplied by 100
Per Game	count	2.3	Simple average
Per 100	rate	18.7	Pace-adjusted
Distances	meters	0-0.91, 0.91-3.05, 3.05-6.75	Native metric
PIR	rating	15.8	Euroleague-specific

Missing Data Handling

Variable	If Missing	Imputation Strategy
Height_cm	Required	Lookup from roster data
Shot zones	Common pre-2015	Use 2PT/3PT splits only
Assist rates	Common pre-2015	Approximate from overall AST
Dunk attempts	Limited availability	Mark as 0 or exclude

Data Export Format

Recommended CSV Structure

CSV

Player,Team,Season,Height_cm,OReb_Rate,DReb_Rate,...,3.05m-3p_FGA_Pct
Mike James,Monaco,2023-2024,185,0.05,0.12,...,25.3
Jan Vesely,Fenerbahce,2023-2024,213,0.18,0.25,...,2.5
...

Column Order (Suggested)

1. Identifiers: Player, Team, Season
2. Physical: Height_cm
3. Rebounding: OReb_Rate, DReb_Rate
4. Playmaking: Assist_Rate, Steal_Rate, Block_Rate, Turnover_Rate
5. Scoring: Points_Per100, Usage_Rate, Player_Efficiency_Rating
6. Free Throws: FTr, FT_Pct
7. Field Goals: FGA_Per100, 2FG_Pct, 3FG_Pct
8. Shot Selection: 2FG_Assist_Rate, 3FG_Assist_Rate, 3FGA_Pct, Corner_3FGA_Pct
9. Shot Zones: Dunk_Attempt_Rate, 0-0.91m_FGA_Pct, 0.91-3.05m_FGA_Pct, 3.05m-3p_FGA_Pct

Ready to Cluster! With all 23 variables in native European metrics, your data is ready for:

- Model-based clustering (Gaussian Mixture Models)
- Player similarity analysis
- Position redefinition
- Lineup effectiveness modeling

No conversions, no approximations, no imperial units! 🎯